

Post-Synthesis Functional Simulation



The screenshot shows a Tcl Console window with a tabbed interface containing 'Tcl Console', 'Messages', and 'Log'. The 'Tcl Console' tab is active, displaying a series of Tcl commands and their outputs. The commands include sourcing a testbench file, setting wave configuration, running a simulation for 1000ns, and checking for top-level signals. The output shows that the simulation completed successfully, with design snapshot 'tb_all_losses_func_synth' loaded and simulation time of 1000ns. Performance metrics for the launch_simulation are provided, including CPU time, elapsed time, memory peak, and gain. Finally, the 'run all' command is executed, showing MSE and CE loss and gradient values.

```
source tb_all_losses.tcl
# set curr_wave [current_wave_config]
# if { [string length $curr_wave] == 0 } {
#   if { [llength [get_objects]] > 0 } {
#     add_wave /
#     set_property needs_save false [current_wave_config]
#   } else {
#     send_msg_id Add_Wave-1 WARNING "No top level signals found. Simulator will start without a wave window. If you want to open a wave window go to 'File->New Waveform'"
#   }
# }
# }
# run 1000ns
==== FULL SYSTEM TEST ====
INFO: [USF-XSim-96] XSim completed. Design snapshot 'tb_all_losses_func_synth' loaded.
INFO: [USF-XSim-97] XSim simulation ran for 1000ns
launch_simulation: Time (s): cpu = 00:00:03 ; elapsed = 00:00:11 . Memory (MB): peak = 3871.969 ; gain = 0.000
run all
MSE Loss: 0x18 | MSE Grad: 0x30
CE Loss: 0x36 | CE Grad: 0xd0
$finish called at time : 10755100 ps : File "C:/posit_loss_project/verilog/tb_all_losses.v" Line 48
```

posit_loss_project - [C:/posit_loss_project/vivado/vivado_build/posit_loss_project.xpr] - Vivado 2025.2

File Edit Flow Tools Reports Window Layout View Run Help Q: Quick Access Synthesis Complete

Flow Navigator SIMULATION - Post-Synthesis Simulation - Functional - sim_1 - tb_all_losses

PROJECT MANAGER

- Settings
- Add Sources
- Language Templates
- IP Catalog

IP INTEGRATOR

- Create Block Design
- Open Block Design
- Generate Block Design

SIMULATION

- Run Simulation

RTL ANALYSIS

- Run Linter
- Open Elaborated Design

SYNTHESIS

- Run Synthesis
- Open Synthesized Design
 - Constraints Wizard
 - Edit Timing Constraints
 - Set Up Debug
 - Open Dataflow Design

Scope

Name	Design Unit	Block Type
tb_all_losses	tb_all_losses	Verilog Mox
dut	top_wrapper	Verilog Mox
gblbl	gblbl	Verilog Mox

Objects

Name	Value	Data Type
clk	1	Logic
rst	0	Logic
start_mse	0	Logic
start_ce	0	Logic
y_hat[7:0]	48	Array
y_true[7:0]	40	Array
p_hat[7:0]	30	Array
label	1	Logic
mse_l[7:0]	18	Array
mse_n[7:0]	30	Array

tb_all_losses.v

tb_all_losses_time_synth.wcfg

Tcl Console

```
send_msg_id Add_Wave-1 WARNING "No top level signals found. Simulator will start without a wave window. If you want to open a wave window go to 'File->New Waveform'
}
}
run 1000ns
=== FULL SYSTEM TEST ===
INFO: [USF-XSim-96] XSim completed. Design snapshot 'tb_all_losses_func_synth' loaded.
INFO: [USF-XSim-97] XSim simulation ran for 1000ns
launch_simulation: Time (s): cpu = 00:00:03 ; elapsed = 00:00:11 . Memory (MB): peak = 3871.969 ; gain = 0.000
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MSE Loss: 0x18 | MSE Grad: 0x30
CE Loss: 0x36 | CE Grad: 0xd0
Finish called at time : 10755100 ps : File "C:/posit_loss_project/verilog/tb_all_losses.v" Line 48
```

Sim Time: 10755100 ps

posit_loss_project - [C:/posit_loss_project/vivado/vivado_build/posit_loss_project.xpr] - Vivado 2025.2

File Edit Flow Tools Reports Window Layout View Run Help Q: Quick Access Synthesis Complete

Flow Navigator SIMULATION - Post-Synthesis Simulation - Functional - sim_1 - tb_all_losses

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y_hat[7:0]	48	Array
y_true[7:0]	40	Array
p_hat[7:0]	30	Array
label	1	Logic
mse_l[7:0]	18	Array
mse_g[7:0]	30	Array
ce_l[7:0]	36	Array
ce_g[7:0]	d0	Array
d1	0	Logic
d2	0	Logic
d3	0	Logic
d4	0	Logic

tb_all_losses.v

tb_all_losses_time_synth.wcfg

Tcl Console

```
source tb_all_losses.tcl
+ eat_mse_were (current wave profile)
```

posit_loss_project - [C:/posit_loss_project/vivado/vivado_build/posit_loss_project.xpr] - Vivado 2025.2

File Edit Flow Tools Reports Window Layout View Run Help

Q: Quick Access

Synthesis Complete

Default Layout

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Simulation - Post-Synthesis Simulation - Functional - sim_1 - tb_all_losses

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tb_all_losses.tb

clk

rst

start_ms

start_ce

y_hat[7:0]

y_true[7:0]

p_hat[7:0]

label

mse_l[7:0]

mse_g[7:0]

ce_l[7:0]

ce_g[7:0]

d1

d2

d3

d4

tb_all_losses.v

tb_all_losses_time.synth.wcfg

Name	Value
clk	1
rst	0
start_ms	0
start_ce	0
y_hat[7:0]	48
y_true[7:0]	40
p_hat[7:0]	30
label	1
mse_l[7:0]	18
mse_g[7:0]	30
ce_l[7:0]	36
ce_g[7:0]	d0
d1	0
d2	n

Tcl Console

Messages

Log

MSE Loss: 0x18 | MSE Grad: 0x30

CE Loss: 0x36 | CE Grad: 0xd0

\$finish called at time : 10755100 ps : File "C:/posit_loss_project/verilog/tb_all_losses.v" Line 48